

The Force Table

Equipment and Setup: Force table with strings and 4 hangers, Set of masses

In this experiment, we will add vectors two ways: With the analytical method, we will calculate the magnitude and direction angle of the sum of two vectors from the magnitude and directions of the vectors we are adding. The force table will provide the experimental method, where we vary hanging weights to find the magnitude and direction angle of the vector sum.

Vector Basics

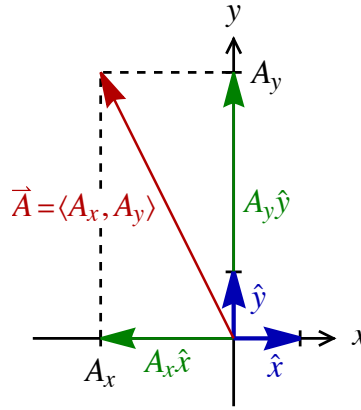
A vector is a physical quantity with both a magnitude and direction. A scalar has just a magnitude. Generally, the magnitude of a vector is a scalar that must be non-negative. A vector can be written in terms of components and the basis unit vectors, or in terms of just components using the angled bracket notation.

$$\vec{A} = A_x \hat{x} + A_y \hat{y} = A_x \hat{i} + A_y \hat{j} = \langle A_x, A_y \rangle$$

The x - and y -components of $\vec{A}$ are A_x and A_y . In the different notations, the basis unit vectors in the x - and y -directions are

$$\hat{x} = \hat{i} = \langle 1, 0 \rangle \text{ and } \hat{y} = \hat{j} = \langle 0, 1 \rangle$$

respectively.



The magnitude of this is A . In two dimensions we can specify a direction by an angle θ , which we will refer to as the direction angle. This angle is unique, modulo 360° ; this means that if two angles differ by a multiple of 360° , they describe the same direction. We use the polar coordinates convention for angles, where an angle is measured counterclockwise from the positive x -axis. The magnitude and direction angle, A and θ , are related to the components, A_x and A_y , just as the polar coordinates, r and θ , are related to the rectangular coordinates, x and y . The components can be found in terms of the magnitude and direction angle:

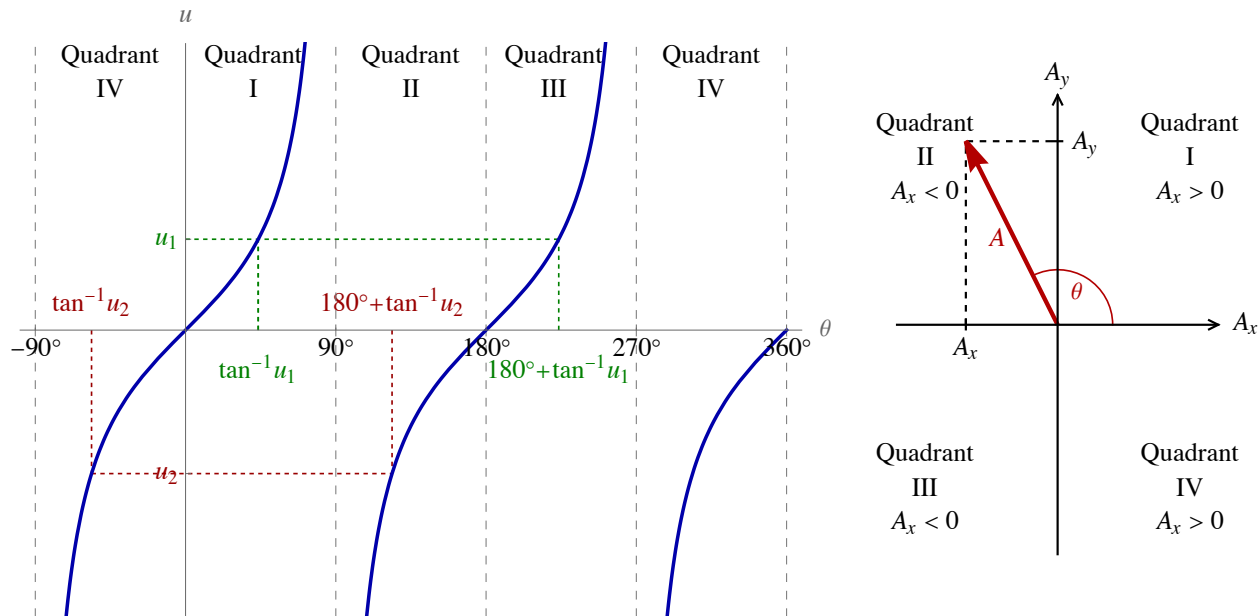
$$A \text{ and } \theta \Rightarrow A_x \text{ and } A_y$$

$$A_x = A \cos \theta \quad \text{and} \quad A_y = A \sin \theta \quad (\text{Equations 1})$$

and the magnitude and direction angle can be found in terms of the components:

$$A_x \text{ and } A_y \Rightarrow A \text{ and } \theta$$
$$A = \sqrt{A_x^2 + A_y^2} \quad \text{and} \quad \theta = \begin{cases} \tan^{-1} \frac{A_y}{A_x} & \text{when } A_x > 0 \\ 180^\circ + \tan^{-1} \frac{A_y}{A_x} & \text{when } A_x < 0 \end{cases} \quad (\text{Equations 2})$$

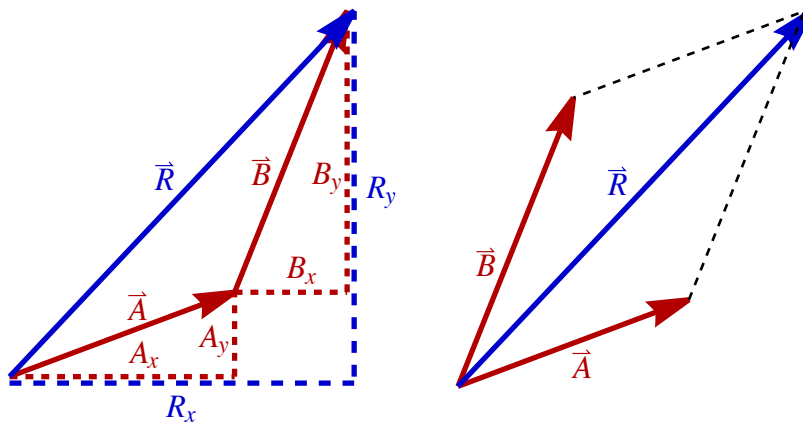
The subtlety in finding the direction angle follows from there being two angles satisfying $\tan \theta = A_y/A_x$, the arctangent of A_y/A_x and 180° added to that arctangent. The arctangent always gives an answer in quadrants I and IV. If the angle is in those quadrants, you use the arctangent. However, if the angle is in quadrants II or III, then you need to take the angle opposite; that is when you add 180° . The sign of A_x determines which is needed. When $A_x > 0$, the vector is in the first or fourth quadrants and when in the second or third quadrants, $A_x < 0$. See the figures below.



When we add vectors, the components of the sum are the sum of the components. If $\vec{A}$ and $\vec{B}$ are vectors then their sum, which we will call the resultant $\vec{R}$, is

$$\vec{A} = \langle A_x, A_y \rangle \text{ and } \vec{B} = \langle B_x, B_y \rangle \text{ then } \vec{R} = \vec{A} + \vec{B} = \langle A_x + B_x, A_y + B_y \rangle = \langle R_x, R_y \rangle$$

Graphically, we add $\vec{A}$ and $\vec{B}$ by putting the tail of $\vec{B}$ at the tip of $\vec{A}$ and the resultant $\vec{R}$ is the vector from the tail of $\vec{A}$ to the tip of $\vec{B}$. Alternatively, we can add the vectors using the parallelogram rule. Put the vector's tails together, complete the parallelogram and draw the vector to the opposite corner.



Forces and the Force Table

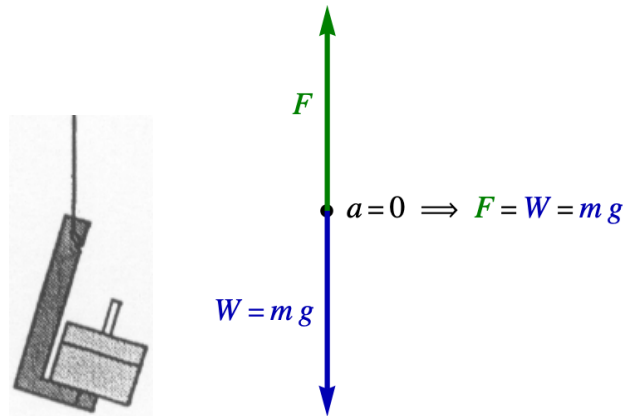
This experiment is about vectors, however the vectors we will add are force vectors. Because the experiment is designed to precede a discussion of forces in the lecture course, we must outline the aspects of

forces needed for the experiment. Since nothing will be moving in this experiment, we have what is known as static equilibrium; in static equilibrium all forces acting on a body must sum to zero: $\sum \vec{F} = \vec{0}$. Note that the zero-vector is the vector with zeros as components: $\vec{0} = \langle 0, 0 \rangle$.

We will hang masses on a string using a hanger with added masses. The total hanging mass is m , the sum of the added mass and the hanger's mass. Weight W is a force; it is the force of gravity acting on an object. In the SI system of units we measure force in newtons N and we measure mass in kilograms kg. The magnitude of the weight is related to the mass by:

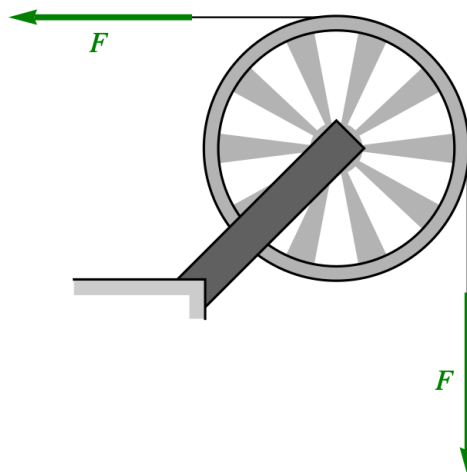
$$W = mg \text{ where } g = 9.80 \text{ m/s}^2$$

For the purposes of this experiment, we will avoid multiplying all masses by 9.80 m/s^2 to get the weight by using $g \cdot \text{kg}$ as our units of force. It follows that the hanging mass in kg corresponds to the same magnitude force in $g \cdot \text{kg}$. For instance, if we have a total hanging mass of 200 grams, that is 0.20 kg and thus corresponds to a weight of $0.20 \text{ } g \cdot \text{kg}$.



Because there is no acceleration, the forces on the hanger must balance.

The other important fact about forces for this experiment is that for an ideal pulley, the string's tension F is the same on either side of the pulley.



An ideal pulley is frictionless and light; this means that the friction in the pulley is negligible, and the mass of the pulley is small compared to the other masses. (In this experiment, since there is no acceleration, the assumption that the pulley is light is not needed.)

Procedure for the Analytical Method

Vector Addition

For Vector Addition 1, 2 and 3, you will be given the magnitude and direction angle of the two (or three) force vectors you are summing. From these magnitudes and direction angles you find the vectors $\vec{F}_1$ and $\vec{F}_2$. Then add these vectors to get the resultant $\vec{R} = \vec{F}_1 + \vec{F}_2$. You will then find the magnitude R and direction angle θ_R of the resultant vector, the sum of the force vectors.

An example will be helpful. Suppose you are given the magnitude and direction angles of $\vec{F}_1$ and $\vec{F}_2$:

$$F_1 = 0.155 \text{ g}\cdot\text{kg}, \theta_1 = 25^\circ, F_2 = 0.275 \text{ g}\cdot\text{kg} \text{ and } \theta_2 = 135^\circ$$

Find the components of the force vectors using (Equations 1).

$$\vec{F}_1 = \langle F_{1x}, F_{1y} \rangle = \langle F_1 \cos \theta_1, F_1 \sin \theta_1 \rangle = \langle 0.14048, 0.06551 \rangle \text{ g}\cdot\text{kg}$$

$$\vec{F}_2 = \langle F_{2x}, F_{2y} \rangle = \langle F_2 \cos \theta_2, F_2 \sin \theta_2 \rangle = \langle -0.19445, 0.19445 \rangle \text{ g}\cdot\text{kg}$$

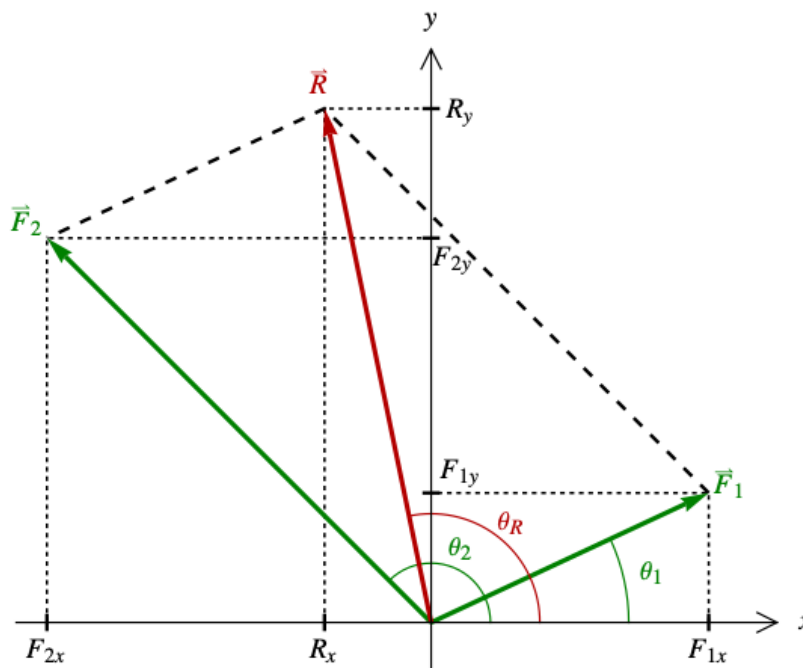
Adding these give the resultant $\vec{R}$ and its components:

$$\vec{R} = \langle R_x, R_y \rangle = \vec{F}_1 + \vec{F}_2 = \langle -0.05398, 0.25996 \rangle \text{ g}\cdot\text{kg}$$

Using (Equations 2) we can find the magnitude and direction angle of the resultant.

$$R = \sqrt{R_x^2 + R_y^2} = 0.266 \text{ g}\cdot\text{kg} \text{ and } \theta_R = 180^\circ + \tan^{-1} \left(\frac{R_y}{R_x} \right) = 101.7^\circ$$

This is the result we are seeking. Note that the 180° was added to the arctangent because $R_x < 0$.



The vector diagram for the worked-out example

For Vector Addition 3, you will be adding three force vectors. Follow the procedure outlined above but now include the third force: $\vec{R} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3$.

Vector Resolution

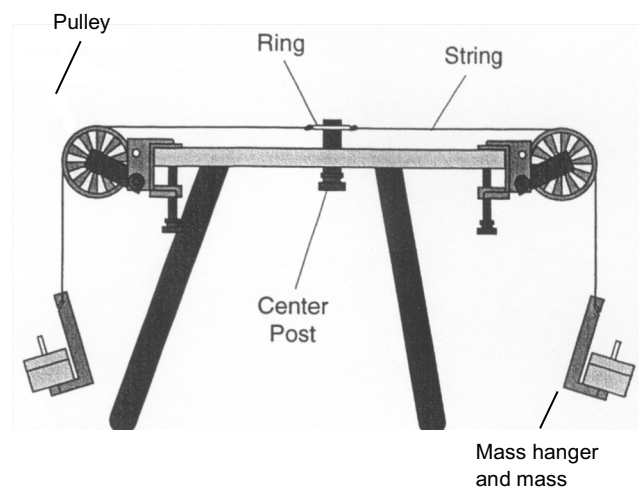
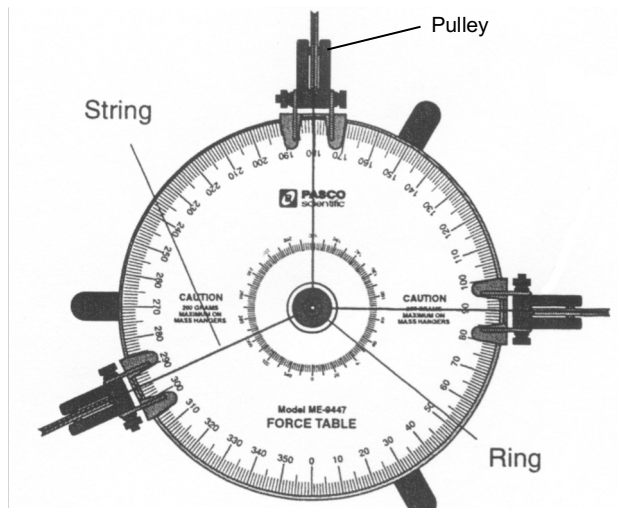
The last two parts of this experiment, Vector Resolution 1 and 2, involve finding the components of vectors. For each case you will be given F and θ , the magnitude and direction angle of some force vector, and you want to find the components F_x and F_y . Calculating these components is a simple application of (Equations 1):

$$F_x = F \cos \theta \quad \text{and} \quad F_y = F \sin \theta$$

Experimental Procedure - the Force Table

Setup

- Obtain a force table (with four pulleys and mass hangers) and a set of masses.
- Looking at the force table from the top, make sure that there is some visible gap between the pulleys and the edge of the force table. All pulleys should be able to rotate freely without rubbing against the edge of the force table. If the pulleys need adjustment, locate the horizontal screw near where the pulley meets edge of the force table. Loosen this screw and tilt the pulley outward until it no longer rubs against the edge of the table.
- Looking at the force table from the side, make sure that all four pulleys are at the same vertical height, so that all the strings lie in a single horizontal plane (or as nearly so as possible). If one or more pulleys need adjustment, use the screw referred to in the previous step to tilt the pulley(s) until all pulleys lie very nearly in a single horizontal plane.
- Hold the ring so that it is centered on the post and look down at the force table from the top. Make sure that all strings pull radially away from the center of the post. From time to time, the strings can slip from side to side along the circumference of the ring so that they are no longer radial. This results in some error in the angle giving the direction of the tensions acting on the ring. If one or more of your strings have slipped off-center, slide them so that all strings pull radially away from the post.



Top View and Side View of Force Table

Vector Addition

To experimentally add two vector, we use the force table. We cannot find the resultant directly. We find the third vector, called the *equilibrant*, that cancels the other two and balances the ring in the force table.

$$\vec{0} = \vec{F}_1 + \vec{F}_2 + \vec{E}$$

It follows that the resultant can then be found from the equilibrant:

$$\vec{R} = -\vec{E}$$

To set $\vec{F}_1$ and $\vec{F}_2$ set the pulleys at angles θ_1 and θ_2 . Then hang the weights F_1 and F_2 from the pulleys. If the force is 0.125 g·kg, then you will hang total mass of 0.125 kg = 125 grams. With those two strings pulling on the ring at the center, take a third string and with your hand find the direction needed to balance the other two forces. This is the direction angle of the equilibrant $\vec{E}$. Set the third pulley at that angle and vary the weights hanging from it until that force balances the other two. The magnitude of the hanging weight in g·kg is the magnitude of the equilibrant, E . But since $\vec{R} = -\vec{E}$ the magnitudes of $\vec{R}$ and $\vec{E}$ are equal, $R = E$. The vectors are in opposite directions, so the direction angle for the resultant θ_R is related to the direction angle of the equilibrant θ_E by $\theta_R = \theta_E \pm 180^\circ$. The magnitude of the resultant is the weight hanging. If the equilibrant is at 250° then the angle of the resultant is 70° .

There are four pulleys and strings but you will only need three for Vector Addition 1 and Vector Addition 2. In those cases, make sure the fourth string is not hanging with any tension; it is best to set that fourth hanger on the force table. Vector Addition 3 will involve adding three force vectors and that will use all pulleys and hangers.

To describe the experimental determination of R and θ using the force table, we will consider the same example given in the analytical method:

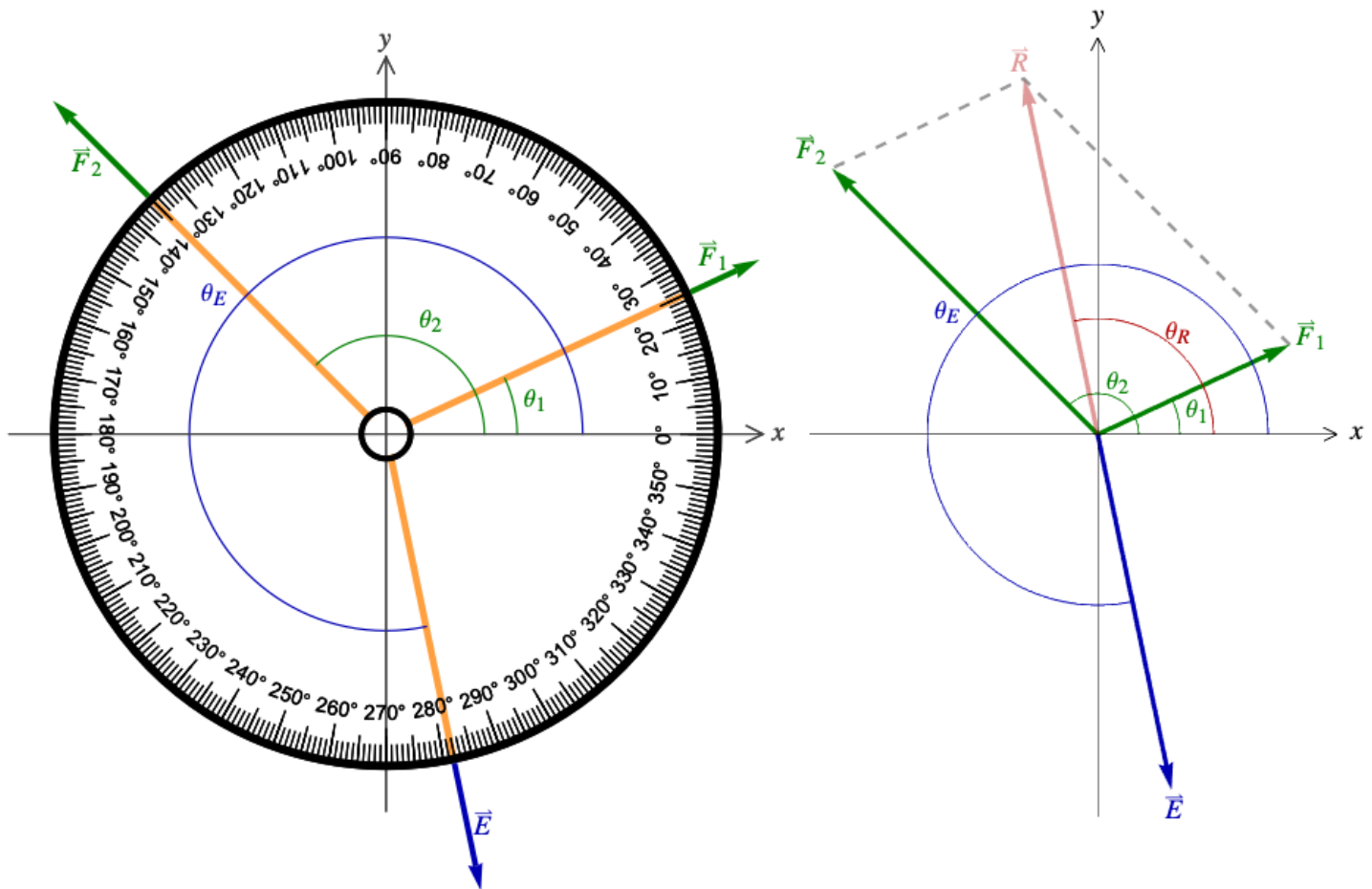
$$F_1 = 0.155 \text{ g}\cdot\text{kg}, \theta_1 = 25^\circ, F_2 = 0.275 \text{ g}\cdot\text{kg} \text{ and } \theta_2 = 135^\circ$$

Over pulleys set at 25° and 135° , hang 0.155 kg = 155 grams and 0.275 kg = 275 grams, respectively. Take a third string and pull in varying directions until you find the direction needed to balance the forces from the other two strings. Suppose this is at 283° ; that then is the direction angle for the equilibrant. Set the third pulley there and start hanging masses over that third pulley until you get it to balance. The total mass hanging gives the magnitude of the equilibrant. Suppose this is 265 grams = 0.265 kg; this then gives the magnitude of $\vec{E}$, the equilibrant, in g·kg:

$$\theta_E = 283^\circ \text{ and } E = 0.265 \text{ g}\cdot\text{kg}$$

Since $\vec{R} = -\vec{E}$ we know that their magnitudes are equal and directions are opposite. This would give the values to record:

$$\theta_R = \theta_E - 180^\circ = 103^\circ \text{ and } R = E = 0.265 \text{ g}\cdot\text{kg}$$



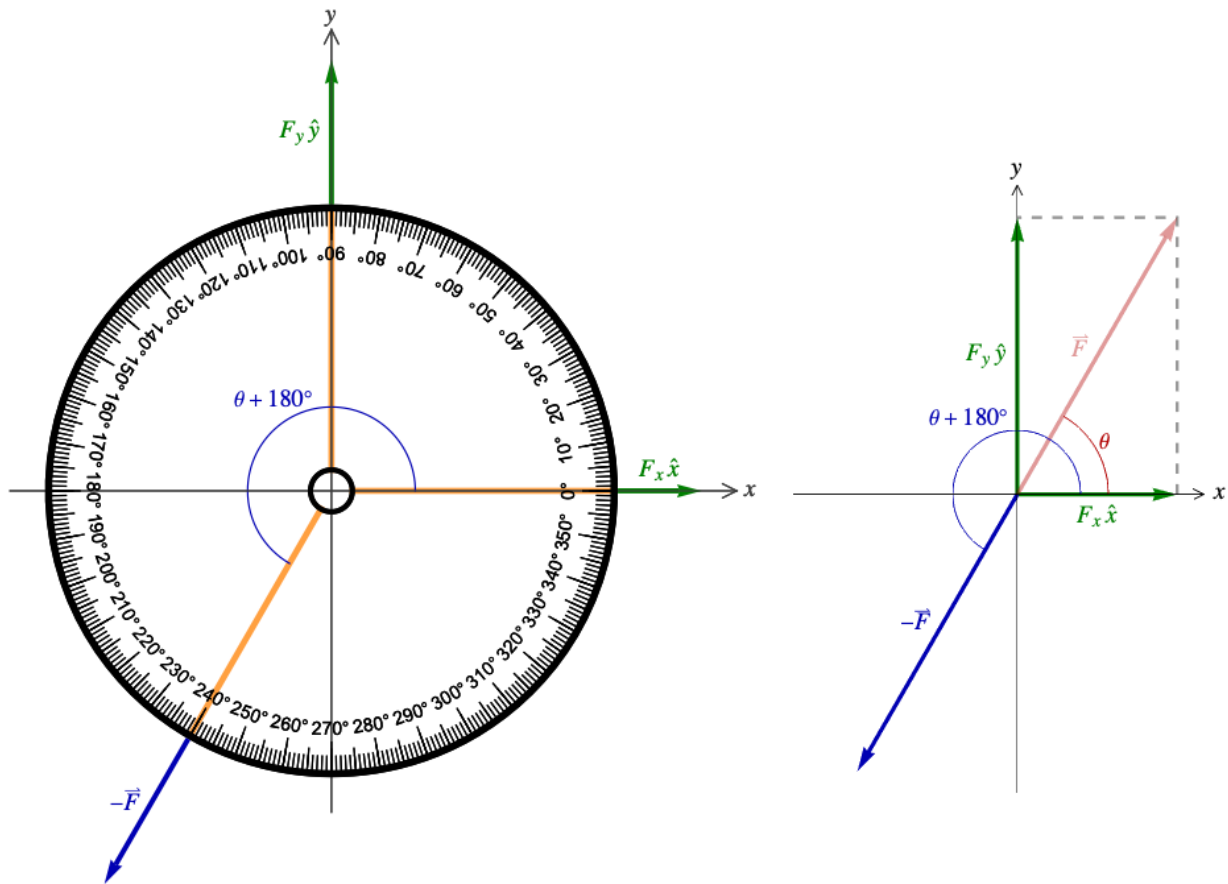
The diagram for the sample example

Vector Resolution

The last two parts of the experiment involve vector resolution, finding the components of a vector using the force table. With Vector Resolution 1, we have an acute angle which makes the procedure simple. To relate this to the Vector Addition part, think of the vector $\vec{F}$, with $F = 0.125 \text{ g}\cdot\text{kg}$ and $\theta = 60^\circ$, as the resultant vector $\vec{R}$; it is the sum of vectors in the x - and y -directions:

$$\vec{F} = F_x \hat{x} + F_y \hat{y}$$

Set three pulleys at 0° , 90° and 240° . We then hang nothing at 60° but hang $0.125 \text{ kg} = 125 \text{ grams}$ opposite to that at 240° . This is $-\vec{F}$ which you should view as the equilibrant. But now we vary the weights hanging at 0° and 90° . Since the angle is acute, both components are positive; what is hanging at 0° is the x -component and what is hanging at 90° is the y -component.



The diagram for Vector Resolution 1

When the angle is not acute we can have negative components. This is the case in Vector Resolution 2. Hang $0.155 \text{ kg} = 155 \text{ grams}$ opposite to $\theta = 125^\circ$, then set other two pulleys along the axes as needed to balance $-\vec{F}$. To get the x -component set the pulley at either 0° or 180° and for the y -component set the pulley at 90° or 270° . The hanging weights will give $|F_x|$ and $|F_y|$. Be careful to include the appropriate signs for F_x and F_y .